

Linux Fundamentals

Week 2

Attendance



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What is Linux?

Definition: An open-source operating system widely used in servers and cybersecurity.

Why Linux?

- Lightweight and customizable.
- Command-line interface (CLI) is powerful for scripting and system control.
- Essential for ethical hacking and security.

Key Features:

- Multi-user system.
- File permissions and process management.

Visual: Diagram showing Linux's role in cybersecurity.



Introduction to the Linux Command Line

Purpose: Allows direct interaction with the OS for tasks like file management, system monitoring, and networking.

Key Commands:

- `ls` – List directory contents.
- `cd` – Change directory.
- `pwd` – Print working directory.
- `chmod` – Change file permissions.
- `cat` – View file contents.
- `mkdir` – Create a directory.
- `rmdir` – Remove an empty directory.
- `rm` – Remove files or directories.
- `touch` – Create an empty file.
- `cp` – Copy files or directories.
- `mv` – Move or rename files or directories.
- `find` – Search for files in a directory hierarchy.
- `grep` – Search for specific patterns in files.

Importance in Cybersecurity:

- Inspecting logs, managing access, testing vulnerabilities.

TryHackMe



Platform: TryHackMe

- **Room:** <https://tryhackme.com/r/room/linuxfundamentalspart1>
- **Activity:** Basic commands, navigating the file system, understanding permissions.
- **Steps:**
 1. Follow instructor guidance for the first sections.
 2. Spend 15-20 minutes completing tasks independently.
 3. Share progress and challenges with the group.

Interactive Activity: File System Scavenger Hunt



Objective: Apply Linux commands to explore a simulated file system and complete challenges.

Setup:

- Make a replit account
- Go to this link <https://replit.com/@josephe1216/Linux-Fundamentals-Activity>

Tasks:

Task 1: Navigate to the `challenge/secrets` directory. What files do you see?

Task 2: Read the contents of `key.txt` using a command.

Task 3: Change permissions for `password.doc` so others can read it.

Task 4: Find the hidden file in the `challenge` directory.

Task 5: Search for "ERROR" in the `access.log`.



Debrief and Key Takeaways

Key Learnings:

- Mastering basic Linux commands is essential for cybersecurity tasks.
- Permissions and file management ensure system security.

Next Steps:

- Explore more Linux commands on TryHackMe or through self-practice.

Real-World Application: Using Linux in penetration testing and server management.

Visual: Checklist or summary graphic.



Q&A Session



Attendance



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Join us for the next sessions



Thursdays: 11AM – 12PM & 2PM – 3PM

Friday: 3PM – 4PM

Monday: 12PM – 1PM



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